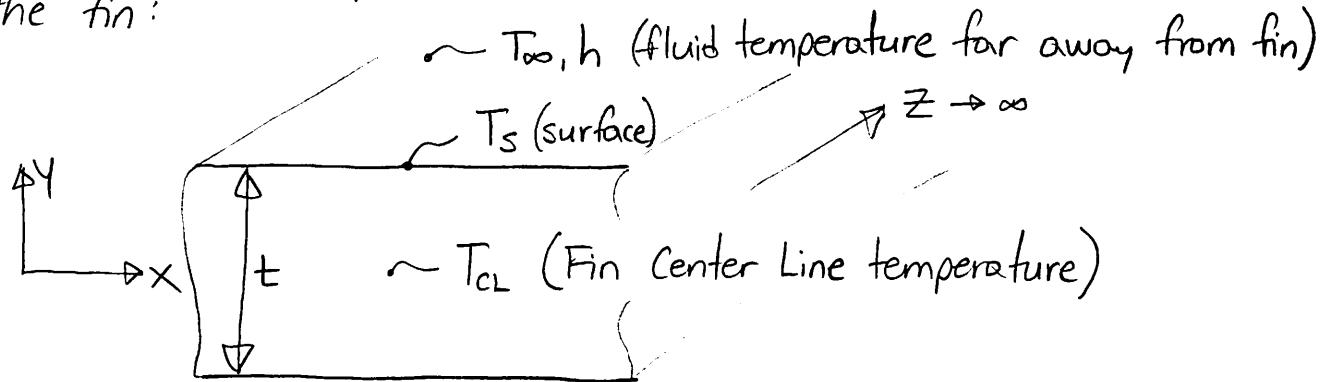


So how do we quantify this? Let's look at a segment of the fin:



Energy balance across the fin (y -direction), we obtain

$$\underbrace{-k \frac{\partial T}{\partial y} (L dx)}_{\text{Conduction}} = \underbrace{h (L dx) (T_s - T_{\infty})}_{\text{Convection}}$$

$$-k \frac{\partial T}{\partial y} = h (T_s - T_{\infty})$$

We can approximate $\frac{\partial T}{\partial y}$ as: $\frac{T_s - T_{CL}}{t/2} = -\frac{(T_{CL} - T_s)}{t/2}$

$$+k \frac{T_{CL} - T_s}{t/2} = h (T_s - T_{\infty})$$

$$\frac{T_{CL} - T_s}{T_s - T_{\infty}} = + \frac{1}{2} \frac{h t}{k} \approx 0.05 \text{ (Set it equal to this)}$$

$$\frac{1}{2} \left(\frac{h t}{k} \right) \leq \frac{1}{20} \Rightarrow \boxed{\frac{h t}{k} \leq \frac{1}{10} \equiv Bi_L} \Rightarrow \text{Biot number}$$

So how do we interpret this approximation or condition?

- 1) Temperature changes across the fin thickness are small compared to those external to the fin

2) Internal thermal resistance to conduction heat transfer across the fin (y-direction) is small compared to the external convective heat transfer resistance. $R_{conv} \gg R_{cond}$

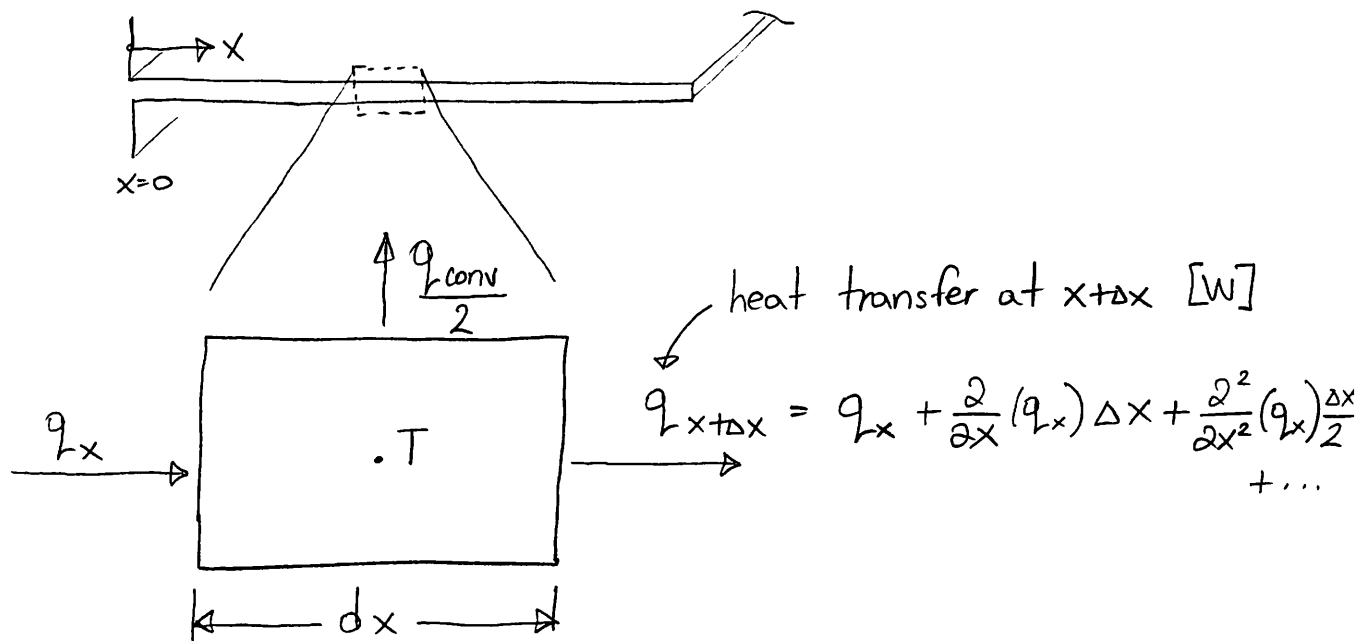
We can double check claim #2 from our previous results:

$$\frac{R_{cond}}{R_{conv}} = \frac{\frac{t}{kA}}{\frac{1}{hA}} = \frac{ht}{k} \ll 1$$

$\underbrace{\quad}_{Bi_t} \ll 1$

So now we can model the heat transfer in the x-direction as 1D $\Rightarrow$ Makes our lives a lot easier!

Let's take a differential fin element:



$$q_{conv} = hPdx(T - T_\infty) \Rightarrow P = \text{perimeter of the fin}$$

$$q_x = -kA \frac{\partial T}{\partial x} \Rightarrow Pdx = \text{surface area/unit length}$$

$Pdx \equiv \text{area for convective heat transfer}$

Let's combine the two equations and do an energy balance on our differential element:

$$E_{in} - E_{out} + E_{gen} = E_{STORED}$$

$\downarrow$ $\downarrow$
 $Q''' = 0$ SS

$$q_x - \left[q_x + \frac{\partial q_x}{\partial x} \Delta x + \frac{1}{2} \frac{\partial^2 q_x}{\partial x^2} \Delta x^2 + \dots \right] - h P \Delta x (T - T_\infty) = 0$$

$$-\frac{\partial q_x}{\partial x} \Delta x - \frac{1}{2} \frac{\partial^2 q_x}{\partial x^2} \Delta x^2 - \dots = h P \Delta x (T - T_\infty)$$

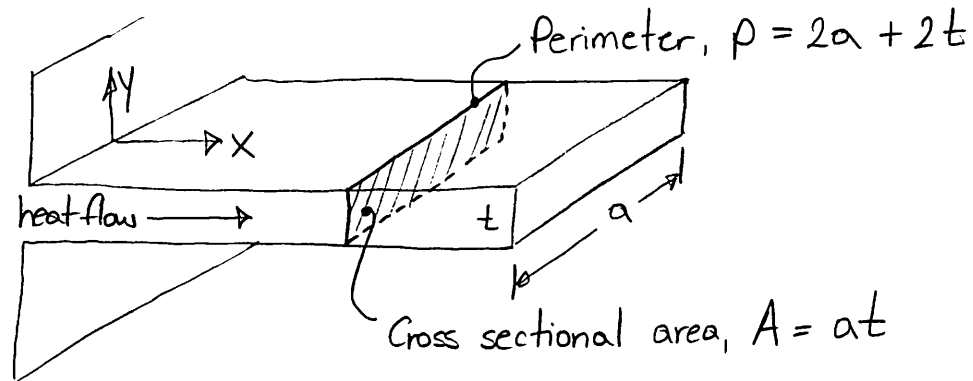
Since we are working with a differential volume: $\Delta x \rightarrow 0$

$$-\frac{\partial q_x}{\partial x} = h P (T - T_\infty) \quad (1)$$

$$q_x = -k A \frac{\partial T}{\partial x} \equiv \text{Fourier's Law} \Rightarrow \text{Back sub into (1)}$$

$$k A \frac{\partial^2 T}{\partial x^2} = h P (T - T_\infty) \quad (2)$$

Note, A is the fin cross sectional area:



To help us solve equation (2), we can assume the following:

$$\underbrace{\theta = T - T_{\infty}}_a ; \quad \frac{\partial \theta}{\partial T} = 1 \Rightarrow \underbrace{\partial \theta = \partial T}_b$$

Convert equation (2) into θ coordinates by back subbing a & b

$$kA \frac{\partial^2 \theta}{\partial x^2} - hP\theta = 0$$

$$\frac{\partial^2 \theta}{\partial x^2} - \frac{hP}{kA} \theta = 0 \Rightarrow \text{Let } \boxed{m^2 = \frac{hP}{kA}}$$

$$\frac{\partial^2 \theta}{\partial x^2} - m^2 \theta = 0 \Rightarrow \text{Linear, second order ODE}$$

Remember from calculus class, if you have an equation of the form:

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = 0$$

$\nwarrow$ $n-1$ derivative $\nwarrow$ a_0 constant

Then you can write a characteristic equation:

$$a_n \lambda^n + a_{n-1} \lambda^{n-1} + \dots + a_1 \lambda' + a_0 = 0 \quad (\text{polynomial})$$

and solve for the roots of the equation: $\lambda = \dots$

For our case:

$$\begin{aligned} \lambda^2 - m^2 &= 0 \\ \lambda^2 &= m^2 \\ \lambda &= \pm m \end{aligned}$$

So our general solution is: $\theta(x) = C_1 e^{mx} + C_2 e^{-mx}$

We can show that this works by the following:

Assume a solution of the form:

$$\begin{aligned}\theta &= C_1 e^{mx} + C_2 e^{-mx} \\ \theta' &= mC_1 e^{mx} - mC_2 e^{-mx} \quad (\text{First derivative, } \frac{\partial \theta}{\partial x}) \\ \theta'' &= m^2 C_1 e^{mx} + m^2 C_2 e^{-mx}\end{aligned}$$

Back substitute into our ODE: $\theta'' - m^2 \theta = 0$

$$\underbrace{m^2(C_1 e^{mx} + C_2 e^{-mx})}_{\theta''} - m^2 \underbrace{(C_1 e^{mx} + C_2 e^{-mx})}_{\theta} = 0$$

$0 = 0 \checkmark$ (Works)

OK, so now back to our problem. How do we solve for C_1 & C_2 (constants). That depends on our B.C.'s

We have a linear second order ODE, so we need 2 B.C.'s

① Infinite Fin ($x \rightarrow \infty$)

Our B.C.'s become:

$$\begin{aligned}\theta(x=0) &= T_b - T_\infty = \theta_b \\ \theta(x \rightarrow \infty) &= T_\infty - T_\infty = 0\end{aligned}$$

As we move out to ∞ , our fin will approach the temperature of the surrounding fluid.

$$\begin{aligned}\theta(x) &= C_1 e^{mx} + C_2 e^{-mx} \\ \theta(\infty) &= 0 = \underbrace{C_1 e^{m(\infty)}}_{C_1=0} + C_2 e^{-m(\infty)}\end{aligned}$$

$C_1 = 0$ for the solution to be valid since $e^{\infty} \rightarrow \infty$